

DEPLOYMENT GUIDE V1

Section 1 – Deployment Governance Framework

1.1 Purpose

Establish governance, controls, approvals, responsibilities, and operational standards governing deployment activities across the Transparent Opportunity Decision Platform (TODP).

1.2 Objectives

- Ensure safe and controlled deployments.
- Minimize production risk.
- Maintain traceability across release activities.
- Support compliance and audit requirements.
- Enable repeatable deployment execution.

1.3 Scope

Applies to:

- Application deployments
- Infrastructure deployments
- Configuration deployments
- Security control deployments
- Database deployments
- Platform service deployments
- Production releases

- Disaster recovery deployments

Covered TODP Modules:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

1.4 Deployment Principles

- Deployment by Approval
- Infrastructure as Code
- Automation First
- Rollback Readiness
- Security by Default
- Traceability by Design
- Separation of Duties
- Production Stability Protection

1.5 Governance Structure

Executive Oversight

→ Product Governance Board

Operational Oversight

→ Platform Operations Committee

Deployment Governance

→ Change Advisory Board (CAB)

Execution Teams

→ DevOps Team

→ Platform Engineering

→ Security Engineering

→ Data Engineering

1.6 Deployment Lifecycle Governance

Development

→ Build

→ Test

→ Security Validation

→ Approval

→ Deployment

→ Verification

→ Monitoring

→ Closure

1.7 Deployment Categories

Category A

Application Deployment

Category B

Infrastructure Deployment

Category C

Security Deployment

Category D

Database Deployment

Category E

Emergency Deployment

1.8 Deployment Control Requirements

All deployments shall include:

- Deployment ID
- Release ID
- Change Reference
- Deployment Owner
- Validation Evidence
- Rollback Plan

- Approval Evidence

1.9 Environment Governance

Development Environment

Integration Environment

QA Environment

UAT Environment

Staging Environment

Production Environment

Environment promotion must follow approved deployment workflows.

1.10 Segregation of Duties

Required separation:

Code Development

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Deployment Approval

Deployment Approval

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Deployment Execution

Deployment Execution

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Audit Validation

1.11 Compliance Requirements

Deployments shall support:

- SOC2 Readiness
- ISO27001 Readiness
- Internal Governance Controls
- Security Architecture V1
- Compliance Package V1

1.12 Audit Requirements

Maintain:

- Deployment Records
- Approval Evidence
- Validation Reports
- Rollback Records
- Deployment Logs

1.13 Metrics

- Deployment Success Rate
- Rollback Frequency
- Deployment Lead Time
- Change Failure Rate

1.14 Ownership

Primary Owner:

DevOps Team

Review Authority:

Platform Operations Committee

Approval Authority:

Change Advisory Board

1.15 Traceability

RTM References:

- DEP-GOV-001
- DEP-GOV-002
- CHG-001
- AUD-DEP-001

1.16 Section Summary

This section establishes deployment governance, approval structures, operational controls, compliance requirements, and audit obligations governing all TODP deployment activities.

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Section 2 – Environment Architecture & Deployment Topology

2.1 Purpose

Define the deployment architecture, environment hierarchy, promotion pathways, and topology standards supporting TODP deployments.

2.2 Objectives

- Standardize deployment environments.
- Reduce deployment inconsistencies.
- Protect production stability.
- Enable controlled software promotion.
- Support operational traceability.

2.3 Environment Architecture

Developer Workstations



Development



Integration



QA



UAT



Staging



Production

2.4 Environment Purposes

Development

- Feature development
- Local testing
- Initial validation

Integration

- Component integration testing
- Service interoperability testing

QA

- Functional testing
- Regression testing

UAT

- Business validation
- Stakeholder approval

Staging

- Production simulation
- Deployment validation

Production

- Live customer workloads
- Operational execution

2.5 Environment Isolation Requirements

Each environment shall maintain:

- Separate credentials
- Separate databases
- Separate infrastructure
- Separate logging scopes
- Separate monitoring scopes

2.6 TODP Deployment Topology

User Layer

API Gateway Layer

Application Services Layer

Opportunity Services

Authority Services

Evidence Services

Retrieval Services

Ranking Services

Recommendation Services

Governance Services

Audit Services

Replay Services

Data Layer

Operational Databases

Audit Repositories

Governance Repositories

Monitoring Layer

Logging Layer

Security Layer

2.7 Production Deployment Zones

Zone A

Application Services

Zone B

Data Services

Zone C

Security Services

Zone D

Monitoring Services

Zone E

Governance Services

2.8 Network Requirements

Production services shall operate within:

- Controlled network boundaries
- Segmented service zones
- Protected ingress controls
- Encrypted communications

2.9 Environment Promotion Rules

Promotion allowed only when:

- Testing completed
- Security validation passed
- Quality gates passed
- Approval received

2.10 Configuration Management

Environment-specific configurations shall be managed through:

- Configuration repositories

- Secure secret stores
- Version-controlled deployment artifacts

2.11 Observability Requirements

Every environment shall provide:

- Metrics
- Logs
- Traces
- Health endpoints

2.12 Disaster Recovery Alignment

Production architecture shall support:

- Recovery deployment
- Backup restoration
- Cross-region recovery
- Operational failover

2.13 Metrics

- Environment Availability
- Promotion Success Rate
- Environment Drift Rate
- Deployment Stability

2.14 Audit Requirements

Maintain:

- Environment Definitions
- Promotion Records
- Configuration Records
- Architecture Reviews

2.15 Traceability

RTM References:

- ENV-001
- ENV-002
- ARCH-DEP-001
- DR-001

2.16 Section Summary

This section defines deployment environments, topology standards, environment isolation requirements, promotion pathways, and production architecture controls supporting TODP deployment operations.

Section 3 – Deployment Prerequisites & Readiness Validation

3.1 Purpose

Define mandatory prerequisites, validation activities, and readiness criteria that must be satisfied before deployment execution.

3.2 Objectives

- Prevent deployment failures.
- Ensure deployment quality.
- Reduce operational risk.
- Verify release readiness.
- Support controlled production releases.

3.3 Scope

Applies to:

- Application releases
- Infrastructure changes
- Security updates
- Database modifications
- Emergency deployments

3.4 Readiness Principles

- Verify Before Deploy
- Test Before Release
- Approve Before Execution
- Validate Before Promotion

3.5 Deployment Readiness Workflow

Build Completion

→ Testing Completion

→ Security Validation

→ Quality Review

→ Approval

→ Deployment Authorization

3.6 Mandatory Deployment Inputs

Required artifacts:

- Release Package
- Deployment Package
- Release Notes
- Test Results
- Security Results
- Rollback Plan
- Deployment Plan

3.7 Build Validation Requirements

Validate:

- Successful compilation
- Dependency verification
- Package integrity
- Artifact generation

3.8 Testing Requirements

Required validation:

- Unit Testing
- Integration Testing
- Regression Testing
- API Testing
- Performance Testing

3.9 Security Validation Requirements

Required checks:

- Vulnerability Scanning
- Dependency Analysis
- Secrets Detection
- Security Policy Validation

3.10 Database Readiness Requirements

Database deployments require:

- Migration scripts
- Rollback scripts
- Data integrity validation
- Backup confirmation

3.11 Infrastructure Readiness Requirements

Validate:

- Infrastructure availability
- Resource capacity
- Network readiness
- Security controls

3.12 Operational Readiness Requirements

Confirm:

- Monitoring coverage
- Alert configuration
- Logging configuration
- Incident response readiness

3.13 Approval Requirements

Required approvals:

- Product Owner
- Engineering Lead
- Operations Lead

Additional approval when required:

- Security Lead
- Governance Lead

3.14 Go/No-Go Review

The Go/No-Go review shall verify:

- Release readiness
- Risk assessment
- Deployment schedule
- Rollback preparedness

3.15 Deployment Authorization Criteria

Deployment may proceed only when:

- All gates pass

- Approvals exist
- Risks accepted
- Documentation complete

3.16 Metrics

- Readiness Compliance Rate
- Failed Readiness Reviews
- Deployment Authorization Success Rate

3.17 Audit Requirements

Maintain:

- Readiness Checklists
- Validation Reports
- Approval Records
- Authorization Records

3.18 Ownership

Primary Owner:

Release Manager

Review Authority:

Platform Operations Committee

Approval Authority:

Change Advisory Board

3.19 Traceability

RTM References:

- READY-001
- READY-002
- REL-001
- DEP-READY-001

3.20 Section Summary

This section establishes deployment readiness controls, validation requirements, approval criteria, authorization processes, and audit obligations required before TODP deployment execution.

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Section 4 – Build, Packaging & Artifact Management

4.1 Purpose

Define standards governing software builds, deployment packaging, artifact creation, storage, integrity validation, and promotion across TODP environments.

4.2 Objectives

- Ensure reproducible builds.
- Maintain artifact integrity.
- Enable traceable deployments.
- Reduce deployment inconsistencies.
- Support audit and compliance requirements.

4.3 Scope

Applies to:

- Application artifacts
- Service packages
- Container images
- Infrastructure artifacts
- Configuration artifacts
- Database migration packages

4.4 Build Principles

- Build Once, Promote Many
- Immutable Artifacts
- Version-Controlled Releases
- Reproducible Execution
- Traceable Packaging

4.5 Build Lifecycle

Source Code

→ Build

→ Test

→ Security Validation

→ Package

→ Store

→ Promote

→ Deploy

4.6 Build Requirements

Every build shall generate:

- Build ID
- Version Identifier
- Commit Reference
- Build Timestamp
- Artifact Manifest

4.7 Versioning Standard

Version Format:

MAJOR.MINOR.PATCH

Example:

1.0.0

1.1.0

1.1.1

4.8 Artifact Types

Category A

Application Packages

Category B

Container Images

Category C

Infrastructure Templates

Category D

Database Migration Packages

Category E

Configuration Packages

4.9 Artifact Repository Requirements

Repositories shall support:

- Version Control
- Access Control
- Integrity Verification
- Retention Policies
- Audit Logging

4.10 Integrity Controls

Artifacts shall be validated through:

- Checksum Verification
- Signature Verification
- Manifest Validation

4.11 Promotion Controls

Artifact promotion requires:

- Successful testing
- Security validation
- Release approval
- Traceable promotion records

4.12 Retention Requirements

Artifacts shall be retained according to:

- Compliance Package V1
- Data Governance V1
- Security Architecture V1

4.13 Metrics

- Build Success Rate
- Artifact Integrity Rate
- Build Duration
- Promotion Success Rate

4.14 Audit Requirements

Maintain:

- Build Logs
- Artifact Records
- Promotion Records
- Integrity Validation Records

4.15 Ownership

Primary Owner:

DevOps Team

Review Authority:

Engineering Leadership

4.16 Traceability

RTM References:

- BUILD-001
- BUILD-002
- ART-001
- AUD-BUILD-001

4.17 Section Summary

This section defines build governance, packaging standards, artifact controls, integrity requirements, promotion procedures, and audit obligations supporting TODP deployments.

DEPLOYMENT GUIDE V1

Section 5 – Infrastructure Deployment Procedures

5.1 Purpose

Establish deployment procedures governing infrastructure provisioning, modification, validation, and promotion across TODP environments.

5.2 Objectives

- Standardize infrastructure deployment.
- Reduce infrastructure risk.
- Maintain configuration consistency.
- Enable scalable platform operations.
- Support infrastructure traceability.

5.3 Scope

Applies to:

- Compute resources
- Networking resources
- Storage resources
- Security infrastructure
- Monitoring infrastructure
- Platform services

5.4 Infrastructure Principles

- Infrastructure as Code
- Immutable Infrastructure
- Automated Provisioning
- Controlled Change
- Security by Default

5.5 Infrastructure Deployment Lifecycle

Infrastructure Definition

→ Validation

→ Approval

→ Provisioning

→ Verification

→ Monitoring

→ Closure

5.6 Infrastructure Components

Application Infrastructure

Database Infrastructure

Network Infrastructure

Security Infrastructure

Monitoring Infrastructure

Governance Infrastructure

Audit Infrastructure

5.7 Deployment Prerequisites

Required before deployment:

- Approved change record
- Capacity validation
- Security review
- Rollback strategy
- Deployment schedule

5.8 Infrastructure Validation

Validate:

- Resource allocation
- Network connectivity
- Security controls
- Monitoring integration
- Backup configuration

5.9 Configuration Controls

Infrastructure configuration shall be:

- Version controlled
- Peer reviewed
- Security reviewed
- Auditable

5.10 Production Infrastructure Deployment

Production deployments require:

- CAB approval
- Operational readiness review
- Deployment authorization
- Post-deployment validation

5.11 Rollback Requirements

Infrastructure rollback plans shall define:

- Rollback trigger
- Rollback owner
- Recovery timeline
- Validation procedure

5.12 Post-Deployment Validation

Verify:

- Resource health
- Connectivity
- Monitoring coverage
- Security posture
- Service availability

5.13 Emergency Infrastructure Changes

Emergency deployments require:

- Incident linkage
- Executive notification
- Post-change review
- Governance validation

5.14 Metrics

- Infrastructure Deployment Success Rate
- Failed Infrastructure Changes
- Rollback Frequency
- Infrastructure Drift Rate

5.15 Audit Requirements

Maintain:

- Deployment Records
- Infrastructure Definitions
- Validation Reports
- Rollback Records

5.16 Ownership

Primary Owner:

Platform Engineering Team

Review Authority:

Platform Operations Committee

Approval Authority:

Change Advisory Board

5.17 Traceability

RTM References:

- INFRA-001
- INFRA-002
- DEP-INFRA-001
- AUD-INFRA-001

5.18 Section Summary

This section defines infrastructure deployment governance, provisioning procedures, validation requirements, rollback controls, and audit obligations supporting TODP platform operations.

DEPLOYMENT GUIDE V1

Section 6 – Application Deployment Procedures

6.1 Purpose

Define deployment procedures governing application services across TODP environments from staging through production.

6.2 Objectives

- Deliver reliable application releases.
- Minimize service disruption.
- Ensure deployment consistency.
- Maintain deployment traceability.
- Support operational resilience.

6.3 Scope

Applies to:

- Opportunity Services
- Authority Services
- Evidence Services
- Retrieval Services
- Ranking Services
- Recommendation Services
- Governance Services
- Audit Services
- Replay Services

6.4 Application Deployment Principles

- Automated Execution
- Controlled Promotion
- Incremental Release
- Continuous Validation
- Rollback Readiness

6.5 Deployment Workflow

Artifact Selection

→ Validation

→ Approval

→ Deployment

→ Verification

→ Monitoring

→ Closure

6.6 Supported Deployment Strategies

Blue-Green Deployment

Rolling Deployment

Canary Deployment

Emergency Deployment

6.7 Deployment Preparation

Verify:

- Release approval
- Artifact availability
- Environment readiness
- Monitoring readiness
- Rollback readiness

6.8 Service Deployment Order

Governance Services

Audit Services

Opportunity Services

Authority Services

Evidence Services

Retrieval Services

Ranking Services

Recommendation Services

Replay Services

The deployment manager may adjust sequencing when operational requirements dictate.

6.9 Configuration Validation

Validate:

- Service configuration
- Secrets configuration
- Environment variables
- Dependency configuration

6.10 Deployment Verification

Verify:

- Service startup
- API availability
- Database connectivity
- Authentication functionality
- Authorization functionality

6.11 Functional Validation

Validate:

Opportunity Processing

Authority Evaluation

Evidence Processing

Retrieval Execution

Ranking Execution

Recommendation Generation

Governance Review

Audit Capture

Replay Processing

6.12 Monitoring Requirements

Post-deployment monitoring shall validate:

- Error rates
- Latency
- Resource utilization

- Service health

6.13 Rollback Criteria

Rollback shall be initiated for:

- Service outage
- Severe degradation
- Security concerns
- Data integrity issues

6.14 Deployment Closure Requirements

Closure requires:

- Validation completed
- Monitoring stable
- Documentation updated
- Approval recorded

6.15 Metrics

- Deployment Success Rate
- Deployment Duration
- Service Stability
- Rollback Rate

6.16 Audit Requirements

Maintain:

- Deployment Logs
- Validation Records
- Approval Evidence
- Rollback Evidence

6.17 Ownership

Primary Owner:

Release Manager

Review Authority:

Platform Operations Committee

Approval Authority:

Change Advisory Board

6.18 Traceability

RTM References:

- APP-DEP-001
- APP-DEP-002
- REL-002

- AUD-APPDEP-001

6.19 Section Summary

This section defines application deployment workflows, deployment strategies, validation requirements, rollback criteria, operational controls, and audit obligations governing TODP service releases.

DEPLOYMENT GUIDE V1

Section 7 – Database Deployment & Migration Management

7.1 Purpose

Define governance, controls, procedures, and validation requirements for database deployments and schema migrations across TODP environments.

7.2 Objectives

- Protect data integrity.
- Minimize migration risk.
- Ensure recoverability.
- Maintain deployment traceability.
- Support operational continuity.

7.3 Scope

Applies to:

- Operational databases
- Governance repositories
- Audit repositories
- Configuration databases
- Metadata stores
- Search indexes

7.4 Database Deployment Principles

- Backup Before Change
- Rollback Before Execution
- Incremental Migration
- Data Integrity First
- Full Traceability

7.5 Database Change Categories

Category A

Schema Changes

Category B

Data Migration Changes

Category C

Performance Optimization Changes

Category D

Security Configuration Changes

Category E

Emergency Database Changes

7.6 Migration Lifecycle

Migration Design

- Review
- Testing
- Backup Validation
- Approval
- Execution
- Verification
- Closure

7.7 Pre-Migration Requirements

Required before execution:

- Approved migration script
- Rollback script
- Backup verification
- Capacity validation
- Change approval

7.8 Data Integrity Validation

Validate:

- Record counts
- Referential integrity
- Index consistency
- Constraint validation
- Query performance

7.9 Production Migration Controls

Production migrations require:

- CAB approval
- Maintenance scheduling
- Recovery validation
- Deployment authorization

7.10 Rollback Requirements

Rollback procedures shall include:

- Recovery steps
- Validation steps
- Ownership assignment
- Recovery timing estimates

7.11 Search Index Deployment Controls

Changes affecting:

- Retrieval indexes
- Ranking indexes
- Evidence indexes

shall undergo:

- Performance testing
- Accuracy validation
- Recovery testing

7.12 Post-Migration Validation

Verify:

- Database availability
- Application functionality
- Data consistency
- Monitoring integration

7.13 Metrics

- Migration Success Rate
- Migration Duration

- Rollback Frequency
- Data Integrity Success Rate

7.14 Audit Requirements

Maintain:

- Migration Scripts
- Approval Records
- Validation Reports
- Rollback Evidence

7.15 Ownership

Primary Owner:

Data Engineering Team

Review Authority:

Platform Operations Committee

Approval Authority:

Change Advisory Board

7.16 Traceability

RTM References:

- DB-DEP-001
- DB-DEP-002
- DATA-001
- AUD-DBDEP-001

7.17 Section Summary

This section defines database deployment governance, migration controls, validation requirements, rollback standards, and audit obligations supporting TODP data platforms.

DEPLOYMENT GUIDE V1

Section 8 – Security Deployment & Validation Procedures

8.1 Purpose

Establish deployment procedures governing security controls, security configurations, access policies, and platform protection mechanisms.

8.2 Objectives

- Maintain secure deployment practices.
- Reduce deployment-related security risk.
- Ensure compliance alignment.
- Preserve security posture.
- Support audit readiness.

8.3 Scope

Applies to:

- IAM configurations
- Authentication services
- Authorization services
- Security policies
- Encryption controls
- Network security controls
- Monitoring security controls

8.4 Security Deployment Principles

- Security by Default
- Least Privilege
- Zero Trust
- Defense in Depth
- Continuous Validation

8.5 Security Deployment Categories

Identity Deployments

Access Control Deployments

Encryption Deployments

Network Security Deployments

Monitoring Security Deployments

Compliance Control Deployments

8.6 Security Deployment Lifecycle

Design

→ Review

→ Security Assessment

→ Approval

→ Deployment

→ Validation

→ Monitoring

8.7 Pre-Deployment Security Requirements

Validate:

- Security review completion
- Risk assessment completion
- Vulnerability assessment results
- Compliance validation

8.8 Access Control Validation

Verify:

- Role assignments
- Privileged access restrictions
- Service account permissions
- Administrative controls

8.9 Encryption Validation

Verify:

- Encryption at rest
- Encryption in transit
- Key management integration
- Certificate validity

8.10 Security Testing Requirements

Conduct:

- Configuration validation
- Vulnerability scanning
- Access testing
- Security control verification

8.11 Production Security Deployment Controls

Production security deployments require:

- Security Lead approval
- CAB approval
- Operational readiness confirmation
- Rollback validation

8.12 Security Monitoring Validation

Verify:

- Alert generation
- Log collection
- Event forwarding
- Detection coverage

8.13 Post-Deployment Security Review

Review:

- Security posture changes
- Access modifications
- Policy compliance
- Monitoring effectiveness

8.14 Metrics

- Security Deployment Success Rate
- Security Control Coverage
- Validation Completion Rate
- Post-Deployment Findings

8.15 Audit Requirements

Maintain:

- Security Reviews
- Validation Reports
- Approval Records
- Security Test Results

8.16 Ownership

Primary Owner:

Security Engineering Team

Review Authority:

Chief Information Security Officer

Approval Authority:

Change Advisory Board

8.17 Traceability

RTM References:

- SEC-DEP-001
- SEC-DEP-002
- SEC-003
- AUD-SECDEP-001

8.18 Section Summary

This section defines deployment procedures, validation standards, approval requirements, and audit controls governing TODP security implementations.

DEPLOYMENT GUIDE V1

Section 9 – Deployment Execution Runbook

9.1 Purpose

Provide a standardized operational runbook governing deployment execution across TODP environments.

9.2 Objectives

- Standardize deployment execution.

- Reduce deployment variability.
- Improve operational reliability.
- Ensure complete deployment traceability.

9.3 Scope

Applies to:

- Standard deployments
- Scheduled deployments
- Emergency deployments
- Application deployments
- Infrastructure deployments

9.4 Runbook Principles

- Execute Approved Changes Only
- Follow Documented Procedures
- Validate Continuously
- Preserve Evidence
- Enable Rapid Recovery

9.5 Deployment Roles

Deployment Manager

Release Manager

Operations Lead

Security Representative

Engineering Lead

Governance Observer

9.6 Deployment Execution Workflow

Step 1

Verify deployment authorization.

Step 2

Verify deployment package integrity.

Step 3

Confirm environment readiness.

Step 4

Verify monitoring availability.

Step 5

Verify rollback readiness.

Step 6

Begin deployment execution.

Step 7

Perform deployment validation.

Step 8

Initiate monitoring review.

Step 9

Confirm service stability.

Step 10

Close deployment.

9.7 Deployment Command Governance

All deployment commands shall be:

- Authorized
- Logged
- Traceable
- Auditable

9.8 Deployment Checkpoints

Checkpoint A

Pre-Deployment Validation

Checkpoint B

Deployment Completion

Checkpoint C

Functional Validation

Checkpoint D

Operational Validation

Checkpoint E

Closure Approval

9.9 Validation Activities

Validate:

- Service health
- API health
- Database connectivity
- Security controls
- Monitoring integration

9.10 Incident Handling During Deployment

Deployment Issue

→ Investigation

→ Impact Assessment

→ Resolution

→ Rollback Decision

9.11 Rollback Execution Workflow

Issue Detection

→ Rollback Authorization

→ Rollback Execution

→ Validation

→ Recovery Confirmation

9.12 Closure Requirements

Deployment closure requires:

- Successful validation
- Stable monitoring period
- Documentation updates
- Approval completion

9.13 Deployment Evidence Requirements

Maintain:

- Deployment logs
- Validation results

- Monitoring screenshots
- Approval records
- Closure records

9.14 Metrics

- Deployment Duration
- Deployment Success Rate
- Validation Completion Rate
- Rollback Frequency

9.15 Audit Requirements

Maintain:

- Execution Records
- Approval Evidence
- Validation Reports
- Operational Logs

9.16 Ownership

Primary Owner:

Deployment Manager

Review Authority:

Platform Operations Committee

Approval Authority:

Change Advisory Board

9.17 Traceability

RTM References:

- RUN-DEP-001
- RUN-DEP-002
- OPS-DEP-001
- AUD-RUNDEP-001

9.18 Section Summary

This section provides the standardized deployment execution runbook, validation checkpoints, rollback procedures, operational controls, evidence requirements, and audit obligations supporting reliable TODP deployments.

DEPLOYMENT GUIDE V1

Section 10 – Post-Deployment Validation & Operational Acceptance

10.1 Purpose

Define validation activities, operational acceptance criteria, and stabilization procedures required following deployment completion.

10.2 Objectives

- Verify deployment success.
- Confirm service integrity.
- Detect deployment defects early.
- Validate business functionality.
- Establish operational acceptance.

10.3 Scope

Applies to:

- Application deployments
- Infrastructure deployments
- Database deployments
- Security deployments
- Emergency deployments

10.4 Validation Principles

- Trust Through Verification
- Validate Before Acceptance
- Evidence-Based Acceptance
- Operational Stability First
- Traceable Outcomes

10.5 Validation Lifecycle

Deployment Completion

- Technical Validation
- Functional Validation
- Security Validation
- Operational Validation
- Acceptance Decision
- Closure

10.6 Technical Validation Requirements

Verify:

- Service startup
- Infrastructure availability
- Database availability
- Network connectivity
- Monitoring integration

10.7 Functional Validation Requirements

Validate:

- Opportunity Processing
- Authority Evaluation
- Evidence Processing
- Retrieval Execution
- Ranking Execution
- Recommendation Generation

- Governance Review Workflow
- Audit Capture Workflow
- Replay Processing Workflow

10.8 Security Validation Requirements

Verify:

- Authentication functionality
- Authorization functionality
- Security logging
- Access controls
- Encryption controls

10.9 Operational Validation Requirements

Confirm:

- Alert generation
- Monitoring visibility
- Dashboard functionality
- Log collection
- Backup operation status

10.10 Stabilization Period

Standard Releases

Minimum stabilization period:

24 Hours

Major Releases

Minimum stabilization period:

72 Hours

Critical Releases

Governance-defined monitoring window

10.11 Acceptance Criteria

Operational acceptance requires:

- Validation completed
- No critical defects
- Monitoring stable
- Security controls operational
- Documentation updated

10.12 Rejection Criteria

Deployment acceptance shall be rejected if:

- Critical functionality fails
- Security controls fail
- Data integrity issues exist
- Recovery procedures fail

10.13 Acceptance Authority

Application Releases

Engineering Lead

Infrastructure Releases

Platform Engineering Lead

Security Releases

Security Lead

Production Acceptance

Operations Lead

10.14 Metrics

- Validation Pass Rate
- Acceptance Rate
- Defect Discovery Rate
- Stabilization Success Rate

10.15 Audit Requirements

Maintain:

- Validation Reports
- Acceptance Records
- Stabilization Reports
- Approval Evidence

10.16 Ownership

Primary Owner:

Operations Lead

Review Authority:

Platform Operations Committee

10.17 Traceability

RTM References:

- VAL-001
- VAL-002
- OPS-VAL-001
- AUD-VAL-001

10.18 Section Summary

This section defines post-deployment validation requirements, operational acceptance criteria, stabilization procedures, governance controls, and audit obligations supporting successful TODP releases.

DEPLOYMENT GUIDE V1

Section 11 – Rollback & Recovery Deployment Procedures

11.1 Purpose

Establish standardized rollback and deployment recovery procedures ensuring rapid restoration of service following deployment failures.

11.2 Objectives

- Minimize service disruption.
- Reduce recovery time.
- Protect data integrity.
- Maintain operational continuity.
- Support deployment resilience.

11.3 Scope

Applies to:

- Application deployments
- Infrastructure deployments

- Database deployments
- Security deployments
- Emergency releases

11.4 Rollback Principles

- Recovery Before Investigation
- Service Stability First
- Controlled Execution
- Full Traceability
- Evidence Preservation

11.5 Rollback Triggers

Critical Service Failure

Severe Performance Degradation

Data Integrity Failure

Security Control Failure

Availability Breach

Governance Approval Failure

11.6 Rollback Lifecycle

Issue Detection

→ Assessment

→ Rollback Decision

→ Authorization

→ Execution

→ Validation

→ Closure

11.7 Rollback Types

Type A

Application Rollback

Type B

Infrastructure Rollback

Type C

Database Rollback

Type D

Configuration Rollback

Type E

Security Rollback

11.8 Authorization Requirements

Rollback authorization may be granted by:

- Operations Lead
- Deployment Manager
- Incident Commander
- Security Lead (security events)

11.9 Application Rollback Procedures

Restore Prior Version

Validate Service Startup

Validate APIs

Validate Monitoring

Confirm Stability

11.10 Infrastructure Rollback Procedures

Restore Approved Baseline

Validate Connectivity

Validate Security Controls

Validate Monitoring

Confirm Operational Readiness

11.11 Database Recovery Procedures

Execute Rollback Scripts

Restore Backup if Required

Validate Integrity

Validate Application Connectivity

Approve Recovery

11.12 Post-Rollback Validation

Verify:

- Service availability
- Functional integrity
- Security controls
- Audit functionality
- Governance functionality

11.13 Recovery Communication Requirements

Notify:

- Operations Teams
- Engineering Teams
- Security Teams
- Governance Teams
- Customers (if required)

11.14 Root Cause Investigation

Mandatory for:

- Production rollback events
- Critical deployment failures
- Security deployment failures

11.15 Metrics

- Rollback Success Rate
- Recovery Duration
- Recovery Validation Success Rate
- Failed Recovery Events

11.16 Audit Requirements

Maintain:

- Rollback Records
- Recovery Evidence
- Root Cause Reports
- Approval Records

11.17 Ownership

Primary Owner:

Incident Commander

Review Authority:

Platform Operations Committee

11.18 Traceability

RTM References:

- RLB-001
- RLB-002
- REC-001
- AUD-RLB-001

11.19 Section Summary

This section defines rollback governance, recovery execution procedures, authorization requirements, validation controls, communication standards, and audit obligations supporting resilient TODP deployments.

DEPLOYMENT GUIDE V1

Section 12 – Deployment Monitoring & Release Observability

12.1 Purpose

Define monitoring, observability, and deployment visibility requirements supporting deployment execution and release stabilization.

12.2 Objectives

- Detect deployment issues rapidly.
- Provide deployment visibility.
- Support operational decision making.
- Reduce deployment risk.
- Enable traceable release monitoring.

12.3 Scope

Applies to:

- Application deployments
- Infrastructure deployments

- Security deployments
- Database deployments
- Production releases

12.4 Observability Principles

- Continuous Visibility
- Early Detection
- Actionable Alerts
- End-to-End Traceability
- Evidence Preservation

12.5 Monitoring Lifecycle

Deployment Start

- Monitoring Activation
- Alert Review
- Validation Review
- Stabilization Monitoring
- Closure

12.6 Monitoring Domains

Application Monitoring

Infrastructure Monitoring

Database Monitoring

Security Monitoring

Governance Monitoring

Audit Monitoring

12.7 Deployment Health Indicators

- Deployment completion status
- Service availability
- Error rate
- API latency
- Resource utilization
- Authentication success rate

12.8 Alert Classification

Critical

High

Medium

Low

Informational

12.9 Deployment Dashboards

Deployment Dashboard

Operations Dashboard

Security Dashboard

Executive Dashboard

Governance Dashboard

12.10 Release Observability Requirements

Every deployment shall provide:

- Metrics
- Logs
- Traces
- Health checks
- Alert coverage

12.11 Monitoring Threshold Management

Thresholds shall be defined for:

- Availability
- Performance
- Security events
- Resource utilization
- Error rates

12.12 Stabilization Monitoring

Monitor during stabilization:

- Availability trends
- Latency trends
- Error trends
- Security events
- Capacity utilization

12.13 Deployment Analytics

Capture:

- Deployment frequency
- Deployment duration
- Failure rate
- Rollback rate
- Service impact

12.14 Metrics

- Deployment MTTD
- Alert Accuracy
- Deployment Stability Score
- Monitoring Coverage

12.15 Audit Requirements

Maintain:

- Monitoring Records
- Alert Records
- Dashboard Evidence
- Stabilization Reports

12.16 Ownership

Primary Owner:

Platform Operations Team

Review Authority:

Platform Operations Committee

12.17 Traceability

RTM References:

- MON-DEP-001
- MON-DEP-002
- OBS-001
- AUD-MONDEP-001

12.18 Section Summary

This section defines deployment monitoring standards, observability requirements, alerting controls, stabilization monitoring procedures, analytics requirements, and audit obligations supporting TODP deployment operations.

DEPLOYMENT GUIDE V1

Section 13 – Emergency Deployment Management

13.1 Purpose

Establish governance, authorization procedures, execution controls, and recovery requirements for emergency deployments required to address critical operational, security, compliance, or business-impacting events.

13.2 Objectives

- Restore critical services rapidly.
- Minimize business disruption.
- Maintain governance oversight.
- Preserve deployment traceability.
- Ensure post-deployment accountability.

13.3 Scope

Applies to:

- Critical production defects
- Security vulnerabilities
- Regulatory remediation requirements
- Service outages
- Infrastructure emergencies
- Data integrity emergencies

13.4 Emergency Deployment Principles

- Controlled Urgency
- Risk-Aware Execution
- Minimum Necessary Change
- Full Traceability
- Mandatory Post-Event Review

13.5 Emergency Deployment Criteria

Emergency deployment may be authorized when:

- Production outage exists
- Security risk is unacceptable
- Regulatory breach risk exists

- Data corruption threatens operations
- Critical customer impact exists

13.6 Emergency Deployment Workflow

Issue Identification

→ Emergency Assessment

→ Risk Review

→ Authorization

→ Deployment

→ Validation

→ Monitoring

→ Post-Implementation Review

13.7 Required Participants

Incident Commander

Operations Lead

Engineering Lead

Security Lead (if applicable)

Governance Representative

13.8 Authorization Requirements

Emergency deployment approval requires:

- Incident reference
- Business justification
- Risk assessment
- Deployment plan
- Rollback plan

13.9 Validation Requirements

Validate:

- Service restoration
- Security controls
- Data integrity
- Monitoring functionality
- Operational readiness

13.10 Communication Requirements

Notify:

- Platform Operations Committee
- Product Leadership
- Security Leadership
- Governance Team

Additional notification shall occur where customer impact exists.

13.11 Post-Implementation Review

Review shall evaluate:

- Deployment effectiveness
- Risk decisions
- Incident linkage
- Recovery outcomes
- Improvement opportunities

13.12 Governance Requirements

All emergency deployments shall undergo:

- CAB retrospective review
- Governance review
- Root cause analysis
- Documentation validation

13.13 Metrics

- Emergency Deployment Volume
- Emergency Deployment Success Rate
- Emergency Rollback Rate

- Time to Restoration

13.14 Audit Requirements

Maintain:

- Emergency Approval Records
- Incident References
- Validation Reports
- Review Documentation

13.15 Ownership

Primary Owner:

Incident Commander

Review Authority:

Platform Operations Committee

Approval Authority:

Emergency Change Authority

13.16 Traceability

RTM References:

- EDEP-001

- EDEP-002
- INC-001
- AUD-EDEP-001

13.17 Section Summary

This section defines emergency deployment governance, authorization requirements, validation controls, communication standards, and audit obligations supporting rapid but controlled remediation activities.

DEPLOYMENT GUIDE V1

Section 14 – Deployment Compliance, Audit & Evidence Management

14.1 Purpose

Establish deployment evidence management, compliance controls, audit support procedures, and deployment record governance requirements.

14.2 Objectives

- Demonstrate deployment compliance.
- Preserve deployment evidence.
- Support internal and external audits.
- Maintain deployment accountability.
- Enable complete deployment traceability.

14.3 Scope

Applies to:

- Production deployments
- Infrastructure deployments
- Security deployments
- Database deployments
- Emergency deployments
- Rollback activities

14.4 Compliance Principles

- Evidence-Based Compliance
- Continuous Audit Readiness
- Traceability by Design
- Immutable Records
- Accountability

14.5 Compliance Alignment

This deployment framework supports:

- SOC2 Readiness
- ISO27001 Readiness
- Security Architecture V1
- Data Governance V1

- Compliance Package V1
- RTM V1 Controls

14.6 Deployment Evidence Categories

Category A

Approval Evidence

Category B

Validation Evidence

Category C

Testing Evidence

Category D

Security Evidence

Category E

Operational Evidence

Category F

Rollback Evidence

14.7 Required Deployment Records

Each deployment shall maintain:

- Deployment ID
- Release ID
- Change ID
- Approval Records
- Validation Reports
- Monitoring Results
- Closure Records

14.8 Evidence Lifecycle

Evidence Creation

- Validation
- Storage
- Protection
- Retention
- Retrieval
- Disposal

14.9 Audit Support Workflow

Audit Request

- Evidence Collection
- Validation
- Submission
- Review
- Closure

14.10 Record Retention Requirements

Deployment records shall follow retention requirements established in:

- Data Governance V1
- Security Architecture V1
- Compliance Package V1

14.11 Evidence Protection Controls

Required controls:

- Access restrictions
- Integrity validation
- Backup protection
- Audit logging
- Retention enforcement

14.12 Compliance Reviews

Monthly:

Deployment Compliance Review

Quarterly:

Deployment Governance Assessment

Annually:

Audit Readiness Assessment

14.13 Metrics

- Evidence Completeness Rate
- Audit Readiness Score
- Record Retrieval Time
- Compliance Exception Count

14.14 Audit Requirements

Maintain:

- Audit Packages
- Evidence Inventories
- Compliance Assessments
- Remediation Records

14.15 Ownership

Primary Owner:

Compliance & Governance Team

Review Authority:

Product Governance Board

14.16 Traceability

RTM References:

- COMP-DEP-001
- COMP-DEP-002
- AUD-001
- AUD-DEP-001

14.17 Section Summary

This section defines deployment compliance controls, audit support procedures, evidence governance requirements, retention standards, and traceability obligations supporting enterprise deployment assurance.

DEPLOYMENT GUIDE V1

Section 15 – Deployment Continuous Improvement & Maturity Management

15.1 Purpose

Establish continuous improvement processes and deployment maturity management practices supporting long-term operational excellence.

15.2 Objectives

- Improve deployment reliability.
- Reduce deployment risk.

- Increase deployment automation.
- Enhance operational efficiency.
- Strengthen governance maturity.

15.3 Scope

Applies to:

- Deployment processes
- Release processes
- Validation processes
- Monitoring processes
- Rollback procedures
- Governance controls

15.4 Continuous Improvement Principles

- Measure Performance
- Learn from Failures
- Automate Repetitive Activities
- Eliminate Operational Waste
- Improve Control Effectiveness

15.5 Improvement Inputs

- Deployment Metrics
- Incident Reviews

- Rollback Reviews
- Audit Findings
- Compliance Assessments
- Security Reviews
- Customer Feedback

15.6 Improvement Lifecycle

Observation

→ Analysis

→ Prioritization

→ Planning

→ Implementation

→ Validation

→ Closure

15.7 Deployment Maturity Domains

Domain A

Deployment Governance

Domain B

Automation

Domain C

Validation

Domain D

Observability

Domain E

Compliance

Domain F

Operational Excellence

15.8 Maturity Levels

Level 1

Ad Hoc

Level 2

Managed

Level 3

Defined

Level 4

Measured

Level 5

Optimized

15.9 Improvement Governance

The Platform Operations Committee shall:

- Review deployment performance
- Prioritize improvements
- Approve maturity objectives
- Track implementation progress

15.10 Annual Deployment Assessment

Assessment shall evaluate:

- Deployment effectiveness
- Automation coverage
- Validation maturity
- Compliance maturity
- Operational resilience

15.11 Deployment Optimization Areas

- CI/CD automation
- Release governance
- Validation automation
- Security automation
- Monitoring automation

15.12 Success Criteria

Improvement initiatives shall demonstrate:

- Risk reduction
- Efficiency gains
- Reliability improvement
- Governance enhancement

15.13 Metrics

- Deployment Maturity Score
- Improvement Completion Rate
- Automation Coverage
- Change Failure Reduction

15.14 Audit Requirements

Maintain:

- Improvement Registers
- Assessment Reports
- Governance Decisions
- Validation Evidence

15.15 Ownership

Primary Owner:

Head of Platform Operations

Review Authority:

Platform Operations Committee

Approval Authority:

Product Governance Board

15.16 Traceability

RTM References:

- IMP-DEP-001
- IMP-DEP-002
- MAT-DEP-001
- AUD-IMPDEP-001

15.17 Section Summary

This section establishes deployment continuous improvement processes, maturity assessment models, governance oversight mechanisms, optimization objectives, and measurement standards supporting long-term deployment excellence.

END OF DEPLOYMENT GUIDE V1